advania

EXECUTIVE & ACADEMIC BRIEFING

Sustainable AI Using Digital Twins

From Apollo 13 to autonomous, carbon-aware twins that observe, orient, decide and act



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WHY IT MATTERS

Rehearse the future before you have to live it

–20–50%

Unplanned downtime

Manufacturing twins with anomaly detection and reinforcement learning

–10–25%

Energy consumption

AI-optimized operations with minimal waste

–30%

Corridor congestion

About 30% in Chattanooga, with signals re-optimized every 4 minutes

–15–30%

HVAC energy

Smart-city building energy management

For boardrooms: measurable gains in downtime, energy and carbon. **For universities:** systems that now decide, not just display.

AGENDA

Six stops from a 1970 rescue to the resilient enterprise

01 **The Origins of Digital Twins**

Apollo 13 → PLM 2002 → NASA 2010

02 **Most Common Application Areas**

Six domains where AI twins deliver

03 **Digital Twins for Smart Cities**

Urban twins as a predictive OODA engine

04 **OODA Loop & Digital Twins**

Observe · Orient · Decide · Act

05 **Sustainable AI — 12 Dimensions**

Greening of AI and AI for greening

06 **Sustainable Business Continuity**

Six dimensions of resilient, green operations

From rescue tool to prediction engine



Apollo 13, 1970: ground replicas and simulators at Mission Control

1970

The precursor: Apollo 13

Telemetry from the damaged craft fed ground simulators to test burns, power and trajectory before radioing the crew.

2002

The architecture: PLM

Dr. Michael Grieves: a physical entity, a virtual entity, and two-way data loops between them.

2010

The name: NASA roadmap

John Vickers coins “Digital Twin”: one physics model per vehicle to predict fatigue and remaining life.

AI twins already pay off across six domains



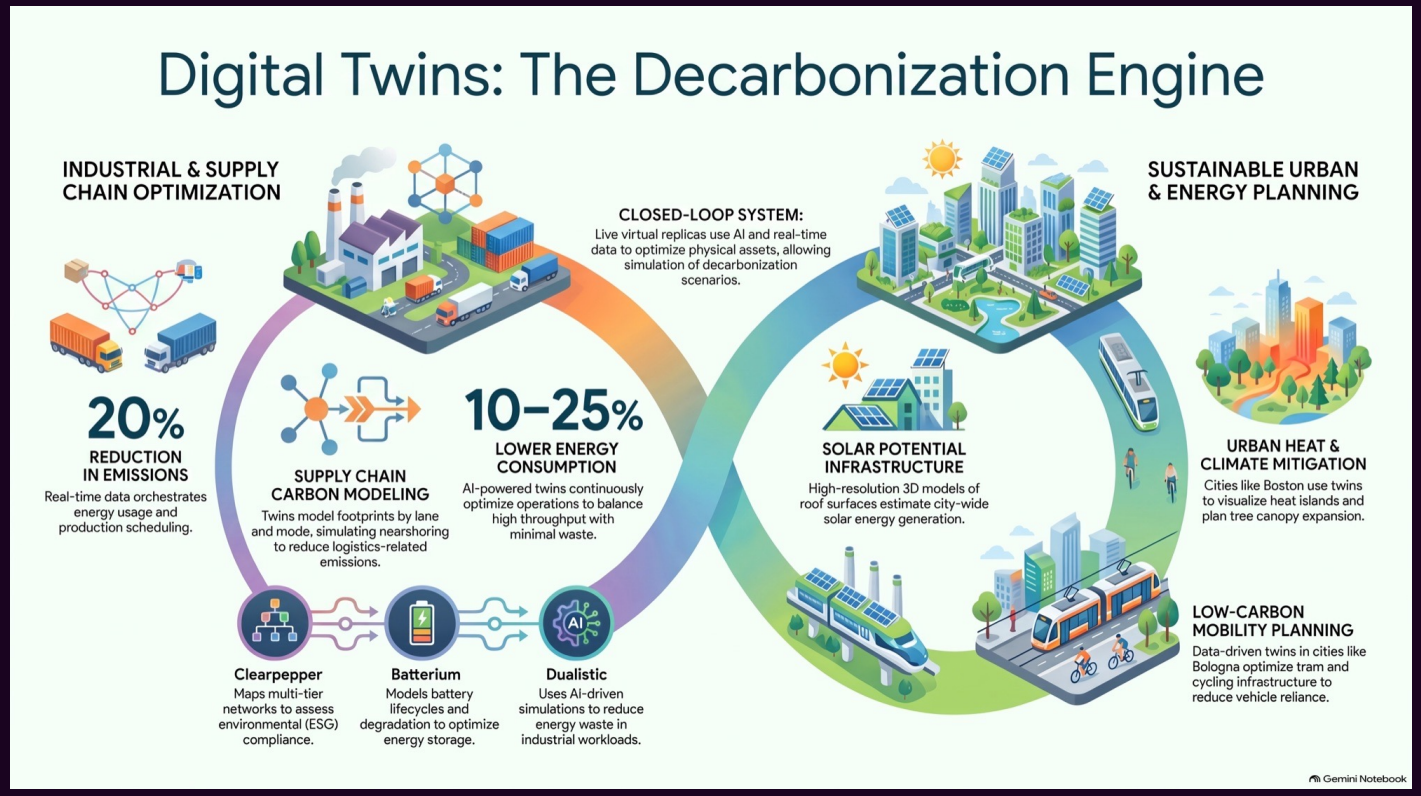
1. **Smart Manufacturing**
   Predictive maintenance, virtual commissioning, RL-tuned lines
2. **Energy & Utilities**
   Turbine health, dynamic grid balancing, pipeline integrity
3. **Supply Chain & Logistics**
   Warehouse orchestration, what-if stress tests, Scope 3 tracking
4. **Healthcare & Life Sciences**
   Patient-specific organ twins, drug discovery, hospital flow
5. **Smart Cities & Infrastructure**
   Traffic and mobility, building energy, disaster simulation
6. **Aerospace & Automotive**
   Engine remaining-life prediction, simulated driving miles

Each domain ties AI to a measurable outcome

Domain	Primary AI technique	Key metric / business outcome
Manufacturing	Anomaly detection, reinforcement learning	Unplanned downtime reduction (typically 20–50%)
Energy & Utilities	Time-series forecasting, physics-informed neural nets (PINNs)	Up to 10–20% boost in efficiency / lower emissions
Supply Chain	Graph neural networks, Monte Carlo simulations	Reduced buffer inventory, faster lead times
Healthcare	Biomechanical modeling, generative diffusion	Lower surgical complication risk, optimized clinical flow
Smart Cities	Computer vision, multi-agent spatial simulation	Reduced transit congestion, 15–30% HVAC energy savings

The pattern turns a technology into an investment case: a clear technique, tied to an outcome you can measure.

Urban twins let cities sense and respond in minutes



An Urban Digital Twin fuses GIS terrain, 3D BIM, municipal IoT sensors and public-service data into one living model.

~30%

Chattanooga, Tennessee

500+ data sources with Oak Ridge National Laboratory; signals re-optimized every 4 minutes.

3D

Virtual Singapore

Wind-flow ventilation, telecom coverage impact and crowd-evacuation simulation.

CO_2

Helsinki 3D+

Building energy history plus solar models to prioritize retrofits block by block.

Four layers power a city twin, three risks hold it back

4
Application & Simulation
Unreal Engine · Cesium · Omniverse · planning sandboxes

3
AI Reasoning & Analytics
Hydrodynamic and microclimate solvers · forecasting · multi-agent RL

2
Data Fusion & Geospatial Storage
GIS + BIM · spatial knowledge graphs · OGC · time-series DBs

1
Ingestion & IoT Telemetry
SCADA · 5G edge · LoRaWAN · Kafka / MQTT streaming

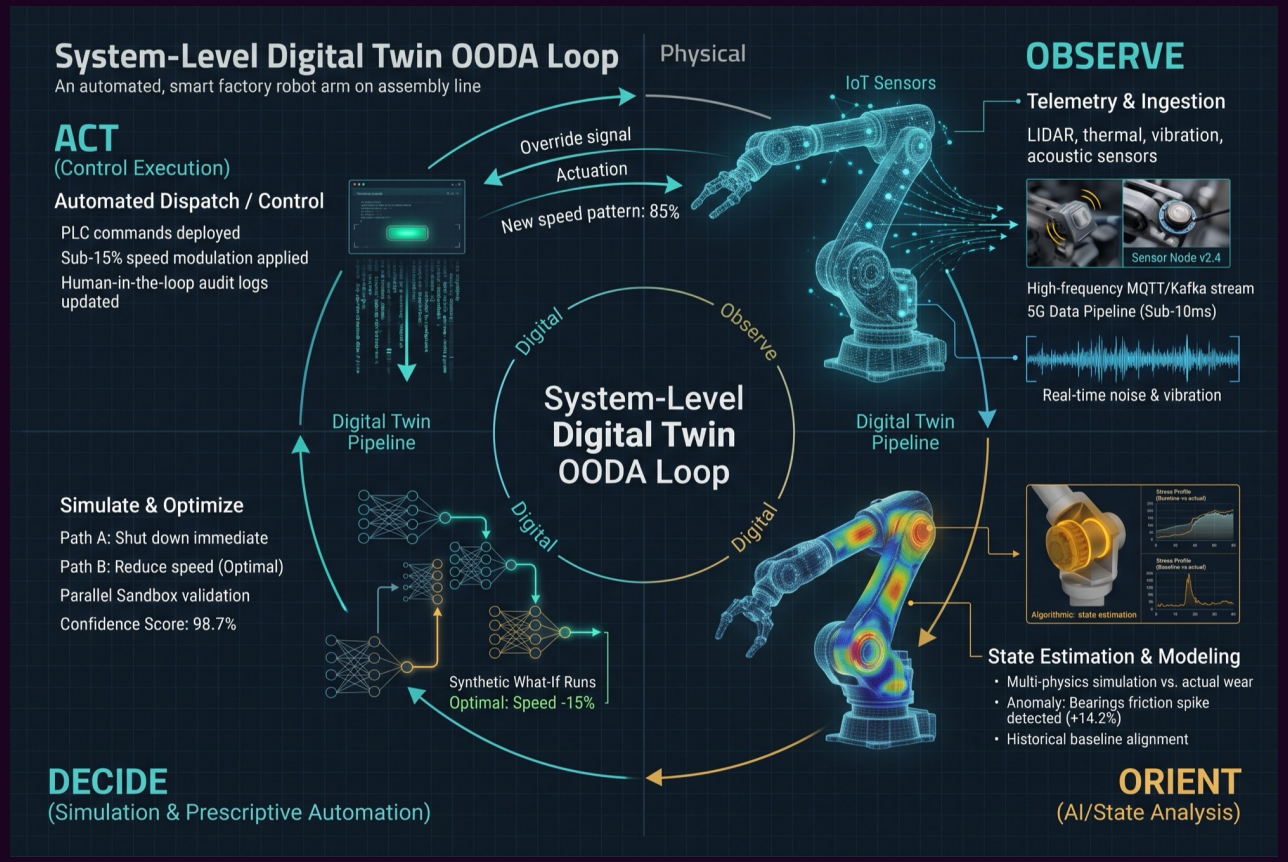
Key operational challenges

Siloed data
Bridging utility SCADA, transit feeds and land registries into one semantic standard (OGC CityGML, Energy ADE).

Citizen privacy
Balancing live computer vision with edge redaction and differential privacy.

Model drift
Keeping the twin in sync with construction, sensor decay and growing trees.

A digital twin is an OODA loop in silicon and sensors



O — Observe

IoT sensors stream vibration, heat and sound in under 10 ms.

O — Orient

Physics + ML tell normal expansion from bearing wear (+14.2% friction).

D — Decide

Parallel what-if simulations pick the best path: slow down 15%.

A — Act

PLC commands go out; the asset responds and the loop restarts.

Twins close the loop in milliseconds, not months

Dimension	Daily life	Technological waves	Digital twins
Clock speed	Hours to weeks	Months to years	Milliseconds to seconds
Primary failure mode	Biased mental models (poor Orient)	Sunk-cost fallacy and organizational inertia	Sensor noise or uncalibrated simulation models
Ultimate objective	Sustainable personal agency and clarity	Market relevance and disruptive advantage	Autonomous resilience and zero-downtime optimization

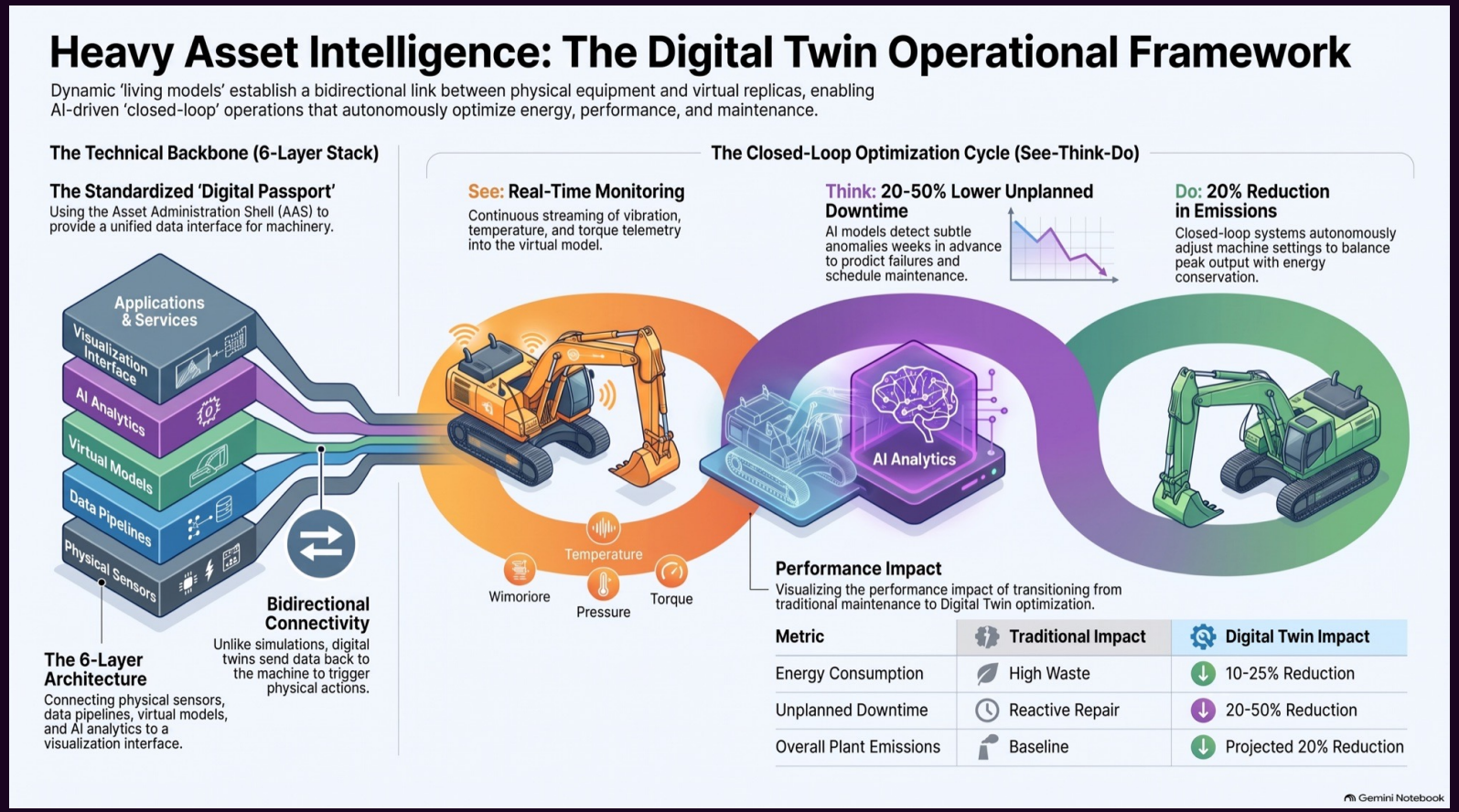
Daily life

Fast personal loops stop small misalignments from compounding into burnout.

Organizations

Modular teams re-orient in days; rigid hierarchies spend months debating the observation.

See, Think, Do pays off in energy and emissions



10–25%

less energy

See ≈ Observe: vibration, temperature and torque streamed live

20–50%

less unplanned downtime

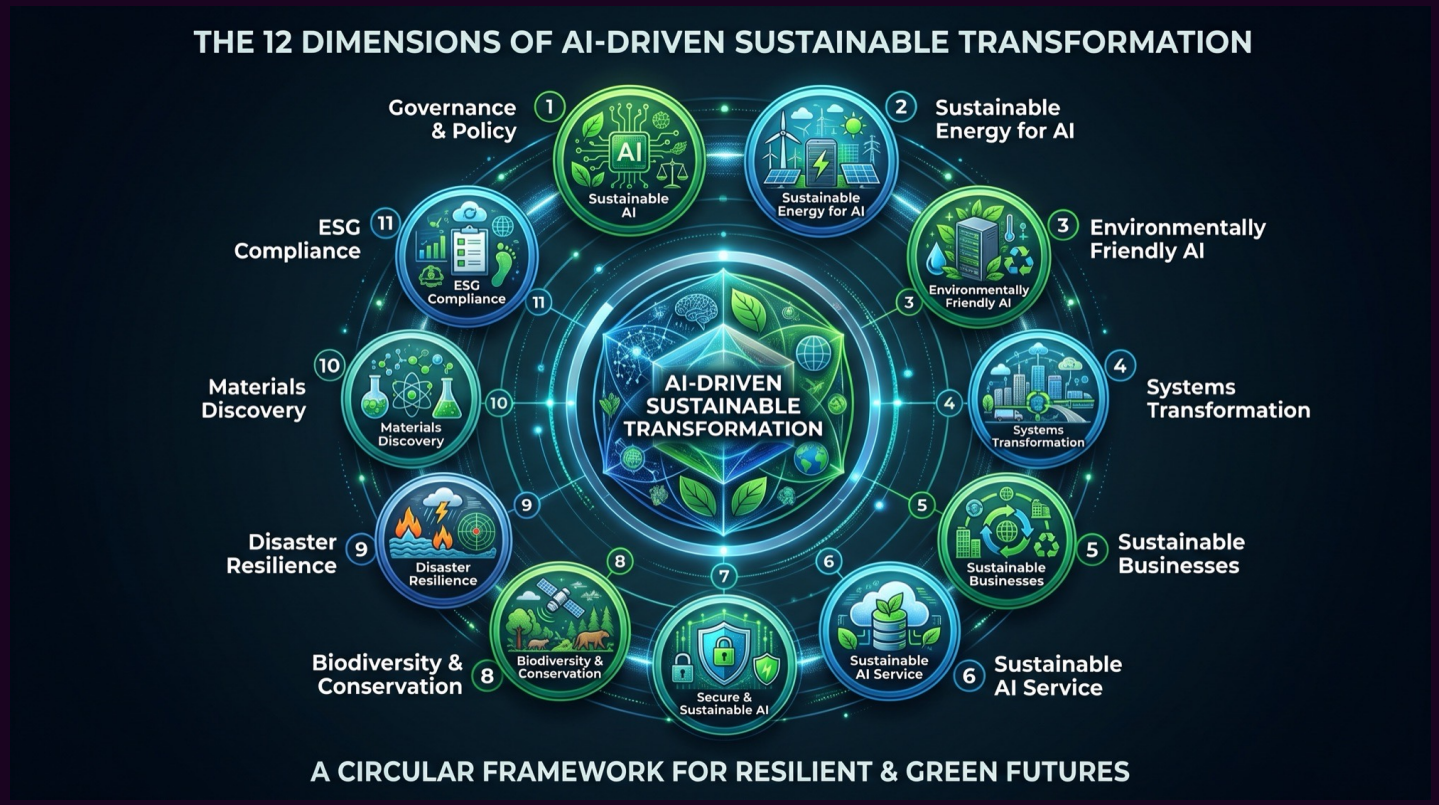
Think ≈ Orient + Decide: anomalies caught weeks ahead

~20%

lower plant emissions

Do ≈ Act: closed-loop control balances output and energy

Sustainable AI has two sides of one coin



Greening of AI

Dimensions 1–3, 6–7

Shrink the footprint of computing: carbon-free power for data centres, efficient models and hardware, right-sized services, secure infrastructure.

AI for greening

Dimensions 4–5, 8–12

Use AI to transform grids, cities, businesses, ecosystems, climate adaptation, materials science, ESG reporting and policy.

Only doing the second side means running a carbon-neutral factory on a coal-powered brain.

Twelve dimensions, held together by governance

1. **Sustainable AI**
   Ethics, equity and long-term lifecycle
2. **Sustainable energy for AI**
   24/7 carbon-free data centres
3. **Green AI**
   Distilled, quantized models; liquid cooling
4. **Systems transformation**
   Grids, traffic and city flows
5. **Sustainable businesses**
   Circular supply chains, Scope 1–3 cuts
6. **Sustainable AI services**
   Right-sizing, carbon-aware routing
7. **Secure & sustainable AI**
   Defence against poisoning and injection
8. **Biodiversity monitoring** ✦
   Satellites and vision guard ecosystems
9. **Disaster resilience** ✦
   Hyper-local early warnings
10. **Materials discovery** ✦
    Graph networks for green chemistry
11. **ESG compliance** ✦
    Automated carbon accounting (CSRD, ISSB)
12. **Governance & policy** ✦
    Disclosure, accountability, reskilling

✦ new dimensions (8–12) extending the original seven

Twins make continuity predictive, lean and green



Resilience has been paid for in carbon. Twins change that bargain.

1. **Predictive foresight**
   Reroute loads before the outage, not after
2. **Lean resilience**
   Stress-test existing capacity instead of idle hot sites
3. **Climate-adaptive operations**
   Model floods, heat and blackouts per site
4. **Circular supply chain**
   Low-carbon rerouting over emergency air freight
5. **Cyber-resilient simulation**
   Rehearse ransomware recovery in a sandbox
6. **Human-centric workspace**
   Safe evacuation, air quality, burnout-free shifts

Resilience no longer has to be paid for in carbon

Dimension	Traditional business continuity	AI digital twin sustainable continuity
Strategy	Reactive incident response and post-event recovery	Predictive, autonomous stress-testing and prevention
Asset redundancy	Carbon-intensive idle hardware and excess stock	Dynamic, software-orchestrated resource sharing
Testing	Disruptive, semi-annual tabletop exercises	Continuous, non-intrusive 24/7 digital simulation
Environmental cost	High-emission emergency pivots (e.g. air freight)	Carbon-optimized routing and microgrid-aware failover

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CLOSE · CALL TO ACTION

Rehearse the future before you live it

In 1970 a copy on the ground brought three people home. Today the stakes are a grid, a city, a supply chain and a climate.

To the executives

Pick one asset or process this quarter and give it a twin with a single measurable goal: downtime, energy or carbon.

To the academics

Help measure the net effect: the carbon AI saves minus the carbon it costs.

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Thank you. Scan to connect on LinkedIn